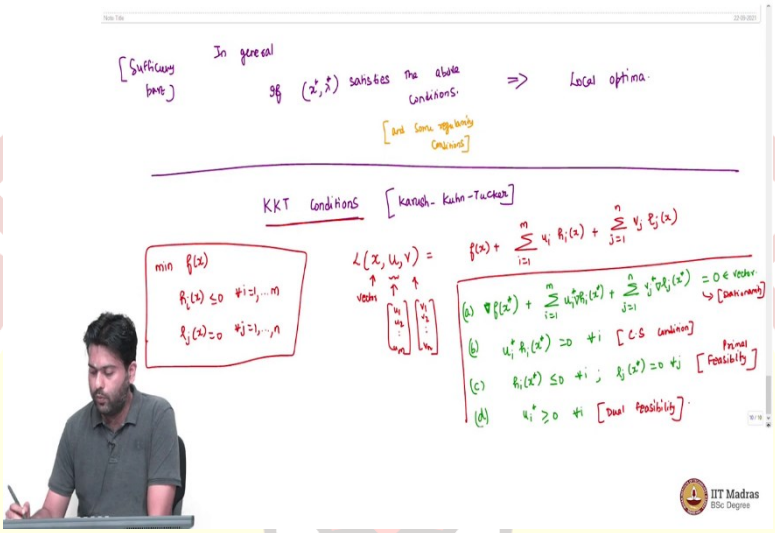


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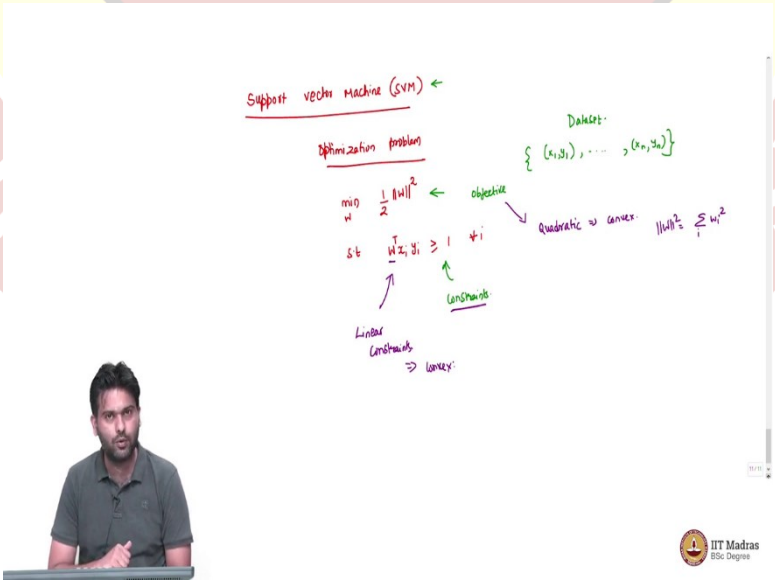
MLF
Professor Arun Rajkumar
Department of Computer Science Engineering
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KKT Conditions Continued

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Now, the next maybe perhaps five minutes, I want to kind of give an overview as to why these things are relevant, so in a machine learning context.

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So, for that, let me look at we will not do the entire exercise of coming up with an algorithm where this is relevant now, but I will just mention that there is this very very fundamental classical algorithm in machine learning called as the support vector machine algorithm. And

you would encounter this algorithm in a future course, on machine learning theory and this algorithm actually can be written as an optimization problem.

What it solves given data can be written as an optimization problem where you have something of this sort. So, the problem would be something like this minimise over w of norm w squared such that I am writing the easiest form of this problem such that w transpose xi yi is greater than or equal to 1 for all i. So, basically, you are given a data set x1 y1 dot xn yn. And I am not trying to motivate why this is the relevant problem here, but I think that would happen in theory course proper machine learning theory course.

But you can see the type of problem from an optimization point of view assuming that this is the problem that is of importance to solve for this given a data set like this, if you want to solve for a particular w which can then be used to make predictions about new data points. So, this is a is an objective that we care to minimise and these are the constraints.

So now, as you can see, this is a constrained optimization problem. What what more is that this objective is half norm w squared. So, norm w squared is simply sum over i wi squared, this is quadratic and quadratic is convex. So, this is a convex function objective quadratic objective which is also convex, and what can you say about the constraints as a function of w.

What kind of functions are these? So, these are linear constraints, linear constraints. So, this is a linear constraint quadratic optimization problem. Now, which means that for such a problem, if I write down the Lagrangian and write the primal problem, because these linear functions are also convex and quadratic functions are also convex, strong duality is going to hold, which means that I can swap the min and max and try to solve the dual problem of this of this particular function.

So, of this particular optimization problem and it turns out that again, we will when you encounter the support vector machine algorithms. So, full formal course, you will realise that, there are several benefits of solving the dual problem as opposed to solving the primal problem. So, in particular, it will, it will help one go from linear models to non-linear models in a very smooth way.

And all these will be covered in the machine learning course. All I wanted to highlight here is that here is a very, very powerful algorithm. That is that is widely used in the machine learning context. And the fundamental idea behind this algorithm depends precisely on converting your

primal problem into a dual problem and deriving the KKT conditions and using that to solve the dual problem and that would give us in this particular problem a lot of insights into the working of this algorithm.

And several algorithms have been developed based on this insights and so on. So, in that sense, constraint optimization problem appear. There is just one example appear a lot in machine learning. And often it is useful to understand the relation between the primal and the dual. Usually, in machine learning the objective functions will be convex.

So duality strong duality will typically hold and you can take advantage of this to solve the dual problems. Which will be typically easier to solve and will also lead to what are called as kernel methods. I will not talk about that in this now. But, so, that is that the point is that there is merit in looking at the dual problem in a machine learning context and support vector machines are an excellent example for that.

So, with that, we conclude the optimization part of this course, just to summarise what all we have seen. So, far in this course, we started with the basic setup of what an optimization problem is, we looked at what does it mean to move in a particular direction, what are good directions to move one that gave us the gradient descent algorithm and then we looked at how directions of the gradients of the objective and the constraint should be related to each other and that gave us the Lagrangian formulation.

We then looked at the class of problems called the convex optimization problems where the unconstrained optimization is easy, I mean it will give you the global optima. And now, we also looked at high level the constraint optimization problem you can go from the primal to the dual view and then there are a, what are called as KKT conditions which happened to be both necessary and sufficient for most convex optimization problems.

And we also saw examples of linear regression for the unconstrained optimization case, which is a machine learning problem and we also have seen support vector machines at least briefly I just wanted to introduce the name here, which is again a quadratic optimization problem with linear constraints, which also can be used using the power of duality theory that we discussed today. With this we will come to an end of the optimization part of this course. I hope all of you enjoyed this part of the course. Thank you.